@prism project

Christophe BAL

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The `@prism` project [1] provides 350 color palettes with native support for CSS, Lua, and LaTeX, to help you create expressive content

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Last changes

Fix.

- Online demo: an auto-update script for the `HTML`, `CSS` and `JavaScript` codes was missing.

Break.

- Palette purification.
  - Deprecated sublist exclusion due to the high overhead of manual reconstruction: return of the palettes `BlueDarkOrange12`, `BlueOrange8`, and `BlueOrange10`.
  - No more visual identical palettes (semi-automated process): `Cubehelix` (`Matplotlib`) was removed, as it is visually identical to `Classic` (`Cubehelix`).

Update.

- Manual.
  - The `Broc`, `GistRainbow`, and `Tofino` palettes have been removed from the cyclic category.
  - The `BlueDarkOrange12`, `BlueOrange8`, and `BlueOrange10` palettes have been added to the sequential category.
  - Added parameters allowing for the reconstruction of palettes omitted due to being shifted variants of existing ones.
  - Added informative appendix describing subpalettes.

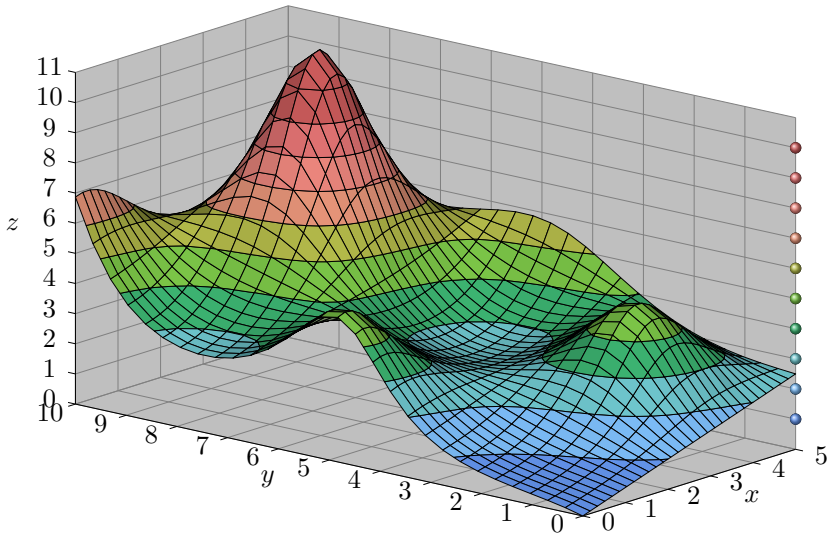
[1] The name comes from "***@·esthetic P·roducts** for **R·epresenting I·nformative S·cientific M·aps***". This name is a double play on words: [1] a prism splits light into an informative spectrum, symbolizing how data are decomposed into meaningful color, and [2] "@" read as "at" indicates where the light meets the prism to be broken down into an informative spectrum.

Contents

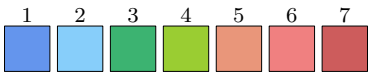
I. Motivations

This project was born out of a desire to enhance `luadraw` with a set of color palettes to easily produce graphics like the 3D plot shown below. Despite the availability of palette generation tools such as `Coolors`, the choice has been made to offer pre-built palettes, because this serves several key purposes for non-designers.

- **Reduced cognitive load:** pre-configuration eliminates design decisions, allowing to focus on content.
- **Consistency:** it becomes easy to ensure visual coherence across documents and projects, regardless of technology.
- **Accessibility compliance:** some pre-configured palettes guarantee colorblind-safe combinations.
- **Perceptual uniformity:** most palettes ensure distinguishable and meaningful colors for data visualization, while others serve artistic purposes.



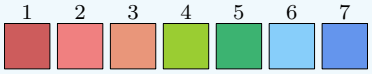
Technically, to generate the graphics in this section, `luadraw` is provided with a finite list of colors and uses linear interpolations to calculate the intermediate color values. In the example above, the finite color palette used is defined as follows.



Using this palette, `luadraw` can produce the following spectrum, allowing us to create the graphic above.

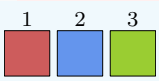
Note.

Using the `Lua` *implementation of* `@prism`*, see the section V-4, via* `luadraw`*, we can create the palettes below made from the previous one named* `'GeoRainbow'`*. Each instruction used is given below each palette.*




```
palCreateFromPal('GeoRainbow', {reverse = true})
```

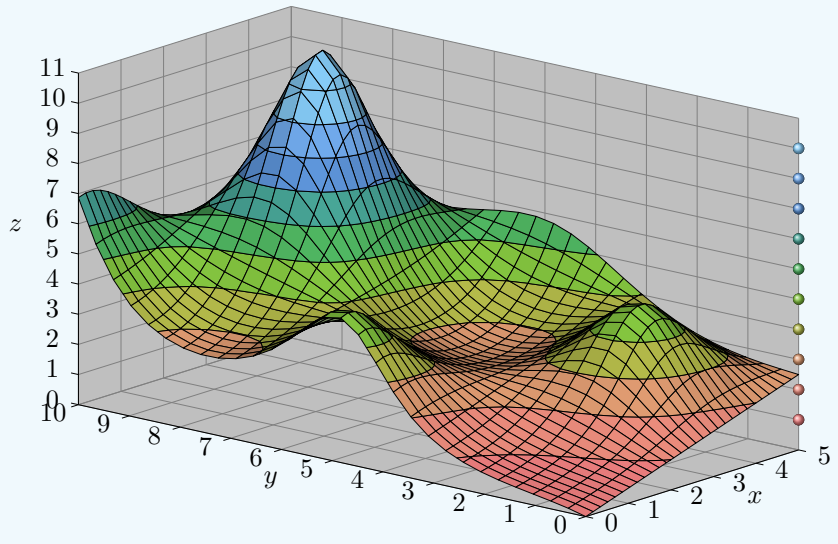

```
palCreateFromPal('GeoRainbow', {shift = 3})
```




```
palCreateFromPal('GeoRainbow', {extract = {7, 1, 4}})
```

This features provide remarkable creative flexibility: with the same surface as before, using the setting `palCreateFromPal('GeoRainbow', {extract = {6, 5, 4, 3, 1, 2}})` *instead of*

`palCreateFromPal('GeoRainbow')`, we change the visual tone, shifting from a seaside feel to a snow-covered world.



II. Where do the color palettes come from?

1. Resources used

In addition to its 10 creations (representing 3 %), `@prism` incorporates 340 palettes from the following projects.

- `Asymptote` is used, but currently offers nothing beyond `Matplotlib` (despite different implementations).
- `CarbonPlan` provides palettes designed for data visualization..
- `CARTOColors` palettes are extracted from `Palettable` project.
- `CMasher` provides palettes designed for data visualization..
- `cmocean` palettes are extracted from `Palettable` project.
- `Colorbrewer` provides professional color palettes for mapping and data visualization.
- `Light and Bartlein` palettes are extracted from `Palettable` project.
- `Matplotlib` compiles color maps from diverse projects.
- `MyCarta` palettes are extracted from `Palettable` project.
- `Plotly` palettes are extracted from `Palettable` project.
- `Scientific Coulour Maps` provides palettes designed for colorblind accessibility.
- `Tableau` palettes are extracted from `Palettable` project.
- `Wes Anderson Palettes` palettes are extracted from `Palettable` project.

Note.

The following versions were used to build the current `@prism` release.

`Asymptote`: 2026-04-05	*`Colorbrewer`: 2021-05-03*
`CarbonPlan`: 2024-01-10	*`Matplotlib`: 2026-04-07*
`CARTOColors`: 2024-09-25	*`Palettable`: 2025-08-23*
`CMasher`: 2026-04-06	*`Scientific Coulour Maps`: 2023-10-05*

2. Palette integration

We retain only palettes that comply with the following rules.

- **No duplicates.** `@prism` maintains a one-to-one mapping[2] between names and palettes.[3]
- **No basic derived versions.** `@prism` omits palettes that can be reconstructed by shifting[4] and/or reversing[5] existing ones, because most `@prism` implementations offer an API to handle these transformations dynamically.
- **Reasonable palette sizes.** Palettes exceeding 512 colors are ignored[6]. This prevents compatibility issues with certain technologies, such as LaTeX, and provides a level of precision far beyond the needs of us mere mortals — or at least, those of us who aren't astronomers.

Note.

Adding new palettes to `@prism` is straightforward (no coding skills required). See section VI-2 to get started with any of the available technologies.

III. Reuse from...

Here are the key points to remember to use palettes offered by projects listed in the section II.

1. Shifted palettes[7] and reversed ones[8] are not included in `@prism`. In rare cases, some palettes are excluded because they appear redundant or are too large. All excluded palettes are listed in Appendix 3 (page 19).
2. `@prism` uses 'semantic' `CamelCase` naming. Therefore, palette names such as `berlin`, `gist_heat` and `Pinkgrey` become `Berlin`, `GistHeat` and `PinkGrey`. To avoid naming collisions, certain palettes have been renamed : a full list of these changes is provided in Appendix 2 (page 17). You can see the full list of names in Appendix 1 (page 14).

Caution.

Most `@prism` implementations add the `pal` prefix to standardized `CamelCase` names. See the section V.

Note.

Most `@prism` implementations provide methods to easily obtain reversed palettes, sub-palettes, and color-shifted palettes. See the section V.

IV. How to choose a palette?

Here are a few tips to help you select your color palette.

1. An `online demo of @prism`[9] provides an intuitive interface for browsing palettes with the following features.
   - **Random discovery:** a 'random' button to explore new palettes.
   - **Category filtering:** quickly narrow down choices by category.
   - **Palette export:** download specific palette files directly.
2. Appendix 5 (page 21) presents all palettes organized by category, including their corresponding discrete definitions for semantic and qualitative categories, and their continuous spectra for other categories.

[2] Some `Matplotlib` palettes are duplicated, likely for historical reasons.
[3] This does not exclude visually similar palettes with distinct names, which are preserved to honor specific design intents. However, if you find palettes with indistinguishable color sets, at least to the naked eye, please report this as a bug.
[4] `Colorcet` offers shifted palettes whose names use an integer suffix.
[5] Most `Matplotlib` colormaps include a reversed version suffixed with `_r`, possibly for performance reasons.
[6] `CMasher` offers the `neutral` palette, which contains 2560 colors
[7] By design, `Colorcet` proposes several shifted palettes.
[8] `Matplotlib` palettes with names ending in `_r` are reversed versions.
[9] `https://projetmbc.github.io/for-writing/`

V. Supported implementations

All implementations are located in the `products` folder. Each implementation provides the following features.

- Modular palette formats (one file per palette).
- Palette definitions in both high-fidelity (original size) and small (currently 40 colors at most) sizes.

Most implementations also feature the API explained below.

- Select specific colors from an existing palette using their indices.
- Shift the palette left (negative value) or right (positive value) by any number of steps.
- Reverse the order of the colors.

Important.

To explain how new palettes can be built, we will refer to the colors in the standard palette as `coul_1`, `coul_2`, etc., and suppose that the extracted indices are `{1, 3, 6, 9}`, the shift used is 1+, and the `reverse` option is enabled. The new palette will then be built sequentially as follows: first `{coul_1, coul_3, coul_6, coul_9}` (extraction), second `{coul_9, coul_1, coul_3, coul_6}` (shift to the right), and finally `{coul_6, coul_3, coul_1, coul_9}` (reverse).

Note.

Extra features are limited to discrete palette operations. For example, color interpolation is not provided, as this is usually handled out of the box by visualization and formatting tools.

1. JSON, the versatile default format

The `JSON` product enables seamless `@prism` palette integration for unsupported programming languages. As shown below, palettes are defined using nested arrays of three-component floats.

```
[
  [0.4980392157, 0.7882352941, 0.4980392157],
  [0.7450980392, 0.6823529412, 0.831372549],
  ...
]
```

2. CSS

Note.

The `HTML`, `CSS`, and `JavaScript` files for product development and demonstration were created using `Claude` and `Gemini` AI assistants.

Each palette color is defined as an individual variable named according to the pattern `--pal<name>-<nb>`, where `<name>` is the standard palette name and `<nb>` is the desired index ranging. Each palette color variable is defined as an `RGB` value using percentage notation. For instance, the following snippet is an excerpt from `palettes-hf/Accent.css`.

```
:root {
    --palAccent-1: rgb(49.803922% 78.823529% 49.803922%);
    --palAccent-2: rgb(74.509804% 68.235294% 83.137255%);
    ...
}
```

The following example illustrates how to generate gradient variables via selective color extraction and custom reordering, while using a standalone color for warning text.

```
:root {
  --transformed-accent-gradient: linear-gradient(
    90deg,
    var(--palAccent-6),
```

```
    var(--palAccent-3),
    var(--palAccent-1)
  );
}

.transformed-accent-gradient {
  background: var(--transformed-accent-gradient);
}

.warning-text {
  color: var(--palAccent-5);
}
```

3. LaTeX

a. Basic use

Accessing a single palette color is straightforward: use `\palUse{<name>}{<index>}` where `<name>` is the standard palette name (without prefix), and `<index>` is the color number. For example, `\palUse{Accent}{8}` is the eighth color of the `Accent` palette, an `xcolor` format color that can be easily used as shown in the following compilable example.

```
\documentclass{article}

% Load the wanted palette.
\usepackage{palettes-hf/Accent}

% Load the palette API.
\usepackage{palapi}

\usepackage{tikz}

\begin{document}

\textcolor{\palUse{Accent}{8}}{\bfseries Colored text.}

Representation of the first ten palette colors.

\begin{tikzpicture}
  \foreach \i in {1,...,10} {
    \fill[\palUse{Accent}{\i}]
      (1.25*\i - 1, 0) rectangle (1.25*\i, 1);
  }
\end{tikzpicture}

\end{document}
```

b. Creating palettes from scratch

Internally, palettes are defined using an array-of-three-floats syntax. For instance, the following snippet is an excerpt from `palettes-hf/Accent.sty`.

```
\palCreateFromRGB{Accent}{
  {0.4980392157, 0.7882352941, 0.4980392157},
  {0.7450980392, 0.6823529412, 0.831372549},
  ...
}
```

For creating new palettes manually, the following high-level commands are available.

1. `\palCreateFromNames` works with a comma separated list of named colors, while `\palCreateFromRGB` creates a palette by entering it as an array-like variable of arrays of three floats.
2. `\palSize{<name>}` returns the palette size (useful for loops, for example).

The following example demonstrates the `\palCreateFromRGB` and `\palCreateFromNames` commands.

```
\usepackage{palapi}
\usepackage[svgnames]{xcolor}

\palCreateFromRGB{MyRGBPal}{
  {0.0, 0.0, 0.0},
  {0.8, 0.2, 0.0},
}

\palCreateFromNames{MyNameUsePal}{
  YellowGreen,
  green!60!black,
}
```

Note.

All built-in palettes are created using the `\palCreateFromRGB` macro.

A lower-level approach is also available through the following commands.

1. `\palNew{<name>}` initializes a new (empty) palette.
2. `\palAddNames{<name>}{<color-using-names>}` appends a color defined with named colors to the palette.
3. `\palAddRGB{<name>}{<r>, <g>, <b>}` appends an `RGB` color to the palette, where `<r>`, `<g>`, and `<b>` are decimal values ranging from 0 to 1.

The following example demonstrates the flexibility offered by these low-level commands.

```
\usepackage{palapi}
\usepackage[svgnames]{xcolor}

\palNew{LowLevelPal}
\palAddNames{LowLevelPal}{IndianRed}
\palAddRGB{LowLevelPal}{0.0, 0.0, 0.0}
\palAddNames{LowLevelPal}{green!60!black}
\palAddRGB{LowLevelPal}{0.8, 0.2, 0.0}
```

c. Creating palettes from existing ones

Building new palettes by transforming existing ones can be achieved using the `\palCreateFromPal` command, which has the signature `\palCreateFromPal{<new-name>}[<options>]{<existing-name>}`. The following example shows how to do this (all options are used).

```
\usepackage{palettes-hf/Accent}
\usepackage{palapi}

\palCreateFromPal{AccentTransformed}[
  extract = {1, 3, 6, 9},
  shift   = 1,
  reverse
]{Accent}
```

Tip.

`\palCreateFromPal{<new-name>}{<existing-name>}` builds a copy of an existing palette.

d. Retrieving the internal definition of a palette

The internally stored definition of a palette named `MyPal`, for example, is `\g_palette_MyPal_seq` which is a LaTeX3 variable (keep in mind the pattern `\g_palette_PaletteName_seq`).

Note.

Variables of type `\g_palette_PaletteName_seq` *are not used internally to retrieve the colors themselves; they are only there for technical reasons related to the development process of new palettes via LaTeX.*

4. Lua

Note.

Initially, the `@prism` *project was created to provide ready-to-use color palettes for* `luadraw`*, a package that greatly facilitates the creation of high-quality 2D and 3D plots using LuaLaTeX and* `TikZ`*.*

a. Basic use

The `Lua` palette variables are named using the prefix `pal`. They are arrays of arrays of three floats (making it straightforward to use a color from a palette). For instance, the following snippet is an excerpt from `palettes-hf/Accent.lua`.

```
palAccent = {
    {0.4980392157, 0.7882352941, 0.4980392157},
    {0.7450980392, 0.6823529412, 0.831372549},
    ...
}
```

b. Creating palettes from existing ones

The `palCreateFromPal` function provides options to build new palettes by transforming existing ones. The following example shows how to do this (all options are used).

```
palAccentTransformed = palCreateFromPal(
    palAccent,
    {
        extract = {2, 5, 8, 9},
        shift   = 1,
        reverse = true
    }
)
```

VI. Contribute via Git

Caution.

Never use the `main` *branch, which is for freezing the latest stable versions of each projects in the mono repository* `https://github.com/projetmbc/for-writing`*.*

1. Complete the translations

Important.

Although we're going to explain how to translate the documentation, it doesn't seem relevant to do so, as English should suffice these days.

The translations are roughly organized as in figure 1 where just the important folders for the translations have been "*opened*".[10] **A little further down, the section VI-1-e explains how to add new translations.**

a. The en folder

This folder, managed by the author of `@prism`, contains files easy to translate even if you're not a coder.

[10] This was the organization on October 26, 2025.

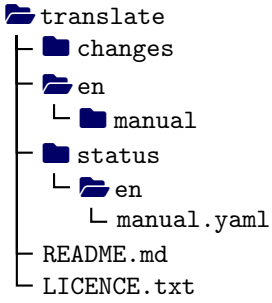


Figure 1: Simplified view of the translation folder

b. The changes folder

This folder is a communication tool where important changes are indicated without dwelling on minor modifications specific to one or more translations.

c. The status folder

This folder is used to keep track of translations from the project's point of view. Everything is done via well-commented `YAML` files, readable by a non-coder.

d. The README.md and LICENCE.txt files

The `LICENCE.txt` file is aptly named, while the `README.md` file takes up in English the important points of what is said in this section about new translations.

e. New translations

Note.

The folder `manual` *is reserved for documentation. It contains* `TEX` *files that can be compiled directly for real-time validation of translations.*

Warning.

Only start from the `en` *folder, as it's the responsibility of the* `@prism` *author.*

Let's say you want to add support for Italian.[11] To do this, you must use `Git` as follows.

1. Via `https://github.com/projetmbc/for-writing/tree/aprism/@prism`, recover the entire project folder. Do not use the `main` branch, which is used to freeze the latest stable versions of all the projects in the mono repository `https://github.com/projetmbc/for-writing`.
2. In the `@prism/contrib/translate` folder, create an `it` copy of the `en` folder, where `it` is the short name of the language documented in the page "*IETF language tag*" from `Wikipedia`.
3. Once the translation is complete in the `it` folder, share it via `https://github.com/projetmbc/for-writing/tree/aprism/@prism` using a classic `git push`.

2. Improving the source code

Participation as a coder is made via the repository `https://github.com/projetmbc/for-writing/tree/aprism/@prism` corresponding to the `@prism` development branch. Here is what you can do, details can be found in the file `https://github.com/projetmbc/for-writing/blob/aprism/@prism/contrib/products/README.md`.

1. Create new palettes within an existing implementation. No coding skills required.
2. Propose a new implementation in your favorite programming language.
3. Combine both approaches.

[11] As mentioned above, there is no real need for the `doc` folder.

VII. History

2.1.0
2026-03-14

Fix.

- Online demo: an auto-update script for the `HTML`, `CSS` and `JavaScript` codes was missing.

Break.

- Palette purification.
  - Deprecated sublist exclusion due to the high overhead of manual reconstruction: return of the palettes `BlueDarkOrange12`, `BlueOrange8`, and `BlueOrange10`.
  - No more visual identical palettes (semi-automated process): `Cubehelix` (`Matplotlib`) was removed, as it is visually identical to `Classic` (`Cubehelix`).

Update.

- Manual.
  - The `Broc`, `GistRainbow`, and `Tofino` palettes have been removed from the cyclic category.
  - The `BlueDarkOrange12`, `BlueOrange8`, and `BlueOrange10` palettes have been added to the sequential category.
  - Added parameters allowing for the reconstruction of palettes omitted due to being shifted variants of existing ones.
  - Added informative appendix describing subpalettes.

2.0.0
2026-02-28

Break.

- Here are the breaking changes in the API.
  - Palette names use 'semantic' camelcase. For example, we use `PinkGrey` and `BWR` (Blue White Red) instead of `Pinkgrey` and `Bwr`, but we keep `RdBu` (RedBlue).
  - `Lua` product: function `getPal` is now `palCreateFromPal`. It expects a palette variable alongside options.
  - Contribution workflow has been fully changed (see below).
  - No more same size for all the palettes (see below).
  - No sublist or shifted palette.
  - The categories have been fully changed (see below).
- Manual.
  - Ignored and renamed palettes have been placed in dedicated appendices.
  - Remove the cluster appendix.

New.

- Here are the new API features.
  - 95 new palettes added.
  - Multiple distribution formats are now available to meet various technical requirements.
    - High-fidelity (unmodified) or normalized (size ≤ 40).
    - One file per palette (no more all palettes in a single file).
  - Categories: the size palette is no longer used since the new API provides named categories. For example, a 3-color smooth palette can be now sequential, qualtitative and/or semantic.
  - Contribution workflow simplified significantly.
    - Data extraction and code generation are now streamlined by instantiating the `PaletteTransformer` class, alongside palette and API builder functions.
    - Standardized showcases now feature simplified graphics.

Update.

- Documentations explain the new API. Here are the changes made.
  - Contribution `README.md` files updated.
  - Manual.
    - Added reference to `CarbonPlan`.
    - Added new category sections with a small description of their meaning.
- The projects listed below serve as a foundation.
  - `Asymptote`
  - `CarbonPlan` (new)
  - `CARTOColors`
  - `CMasher` (new)
  - `Colorbrewer`
  - `Colormaps` (new, just for category automation)
  - `Matplotlib`
  - `NCAR NCL color tables` (new, partially supported for now)
  - `Palettable`
  - `Scientific Coulour Maps`

1.3.0
2026-01-01

Fix.

- Lua product: the `use-std.tex` and `use-dark.tex` codes have been fixed.

Break.

- luadraw product becomes Lua product.
- Lua product: the dict-like variable palNames has been removed. Therefore, the responsibility of adding this specific variable will lie with luadraw.

New.

- New CSS and LaTeX products have been coded.
- The Palettable palettes have been added.
- Three semantic palettes `TdocBW`, `TdocCol` and `TdocDark` have been created to be used by the tutodoc class (LaTeX creation process used).
- Added an online demo of @prism (the CSS product showcase is used).

Update.

- Manual.
  - Added lists of renamed and removed palettes.
  - Added two appendixes, one with all palettes grouped by user-friendly categories, and another showing similar palettes.
  - Added explanations about the utility of palettes for non-designers.
  - Added explanations about the minimal common optional sets of extra features implemented.
- Better palette references.
- Better showcases.
  - Several category PDF files are proposed (one PDF per category and theme).
  - 2D chart and Delaunay triangulation added.

1.2.1
2025-11-09

Fix.

- Equal palettes: the floating point equality uses now a correct tolerance.

Break.

- Palettes: the extra `Greys` has been removed (it is equal to `Grays`).

New.

- Similar palettes: two PDF files show similar palettes in standard and black modes (semi-automated process used).

Update.

- luadraw product: the associative array palNames has been added for compatibility reasons with the luadraw package.
- `BlindFish` palette: the last color variation has been made smoother (luadraw process used).

1.2.0
2025-10-29

Break.

- Palettes: all final palettes now consist of 10 colors.
- luadraw products: the getPal dictionary array has been converted into a function accepting string palette names (with or without pal prefix). See below.

New.

- Palettes.
  - Added `Lemon` and `ShiftRainbow` palettes (luadraw creation process used).
  - Added 37 palettes from the Scientific Coulour Maps project.
- luadraw product: accessing a palette and creating new ones can be made using the getPal function which has an optional argument options (dict-like array) with the following keys and their values.
  - extract: a list of non-zero integers used to extract specific colors from the palette (the order is preserved).
  - reverse: a boolean value indicating whether to reverse the palette color order (false by default).
  - shift: an integer value for applying a circular color shift to the palette.
- Documentations
  - Added English PDF manual.
  - Showcase: two PDF files demonstrate the use of each palette (white and dark modes).

1.1.0
2025-10-14

Break.

- Duplicate palettes and those that are reverse of others are ignored (strict equalities only).

New.

- New palettes added: `BurningGrass`, `GeoRainbow` and `PastelRainbow` (`luadraw` creation process used).
- The `luadraw` palette product has a new dictionary like variable `getPal` to access a palette using its name (as a string variable).

Update.

- Palette contributions: in the mandatory `extend.py` file, the `build_code` function must work with the dictionary of all the palettes, and manage a credit to the `@prism` project.

1.0.0
2025-10-11

First public version of the project.

Appendix 1 – All palette names

Important.

The palette names in this appendix are standard, but some `@prism` implementations add a specific prefix.

A

Accent
Acton
AfmHot
AgGrnYl
AgSunset
Algae
Alphabet
Amber
Amethyst
Amp
Antique
Apple
Aquatic1
Aquatic2
Aquatic3
Arctic
ArmyRose
Autumn

B

Balance
Bam
Bamako
BamO
Batlow
BatlowK
BatlowW
Berlin
Bilbao
Binary
Blackbody
BlindFish
BlueDarkOrange12
BlueDarkOrange18
BlueDarkRed12
BlueDarkRed18
BlueGray
BlueGreen
BlueGrey
BlueOrange8
BlueOrange10
BlueOrange12
BlueOrangeRed
BlueRed
BlueRedPLY
Blues
Blues7
Blues10
BluesCP
BluGrn
BluYl
Bold
Bone
BrBG
BRG
Broc
BrocO
BrownBlue10
BrownBlue12
BrwnYl
Bubblegum
Buda
BuGn
Bukavu
BuPu
Burg
BurgYl
BurningGrass
BWR

C

Cavalcanti
Chevalier
Chroma
Cividis
Classic
CMRmap
ColorBlind
Cool
CoolCP
CoolWarm
Copper
CopperCM
Cork
CorkO
Cosmic
Cube1
Cubehelix1
Cubehelix2
Cubehelix3
CubeYF
Curl

D

Darjeeling1
Darjeeling2
Darjeeling3
Darjeeling4
Dark2
Dark24
DarkMint
Davos
Deep
Delta
Dense
Devon
Dusk

E

Earth
EarthCP
Eclipse
Electric
Ember
Emerald
Emergency
Emrld

F

Fall
FallCM
FantasticFox1
FantasticFox2
Fes
Fire
Flag
Flamingo
Freeze
Fusion

G

G10
GasFlame
Gem
GeoRainbow
Geyser
GhostLight
GistEarth

GistGray
GistHeat
GistNcar
GistRainbow
GistStern
Glasgow
GnBu
Gnuplot
Gnuplot2
Gothic
GrandBudapest1
GrandBudapest2
GrandBudapest3
GrandBudapest4
GrandBudapest5
Gray
GrayC
GreenGrey
GreenMagenta
GreenOrange
Greens
GreensCP
Greys
GreysCP
Guppy

H

Haline
Hawaii
Heart
Holly
Horizon
Hot
HotPLY
HSV

I

Ice
IceBurn
Imola
Inferno
Infinity
IsleOfDogs1
IsleOfDogs2
IsleOfDogs3

J

Jet
JetPLY
JimSpecial
Jungle

L

Lajolla
Lapaz
Lavender
Lemon
Light24
Lilac
LinearL
Lipari
Lisbon

M

Magenta
Magma
Managua
Margot1
Margot2
Margot3
Matter
Mendl
MidGray
Mint
Moonrise1
Moonrise2
Moonrise3
Moonrise4
Moonrise5
Moonrise6
Moonrise7

N

Navia
NaviaW
Neon
NipySpectral
Nuclear
Nuuk

O

Ocean
OceanCM
OkabeIto
Oleron
OrangeBlue
OrangeGrey
Oranges
OrangesCP
OrRd
OrYel
Oslo
Oxy

P

Paired
Pastel
Pastel1
Pastel2
PastelRainbow
Peach
Pepper
PerceptualRainbow
Petroff6
Petroff8
Petroff10
Phase
Picnic
Pink
PinkGreen
PinkGrey
Pinks
PinkYl
PiYG
Plasma
Plotly
Portland
PrGn
Pride
Prinsenvlag
Prism
PrismCC
PuBu
PuBuGn
PuOr
PuRd
Purp
Purple
PurpleGray
PurpleGrey
Purples
PurplesCP
PurpOr

R

Rainbow
RainbowCP
RainbowPLY
Rainforest

RdBu
RdGy
RdPu
RdYlBu
RdYlGn
Red
RedGrey
RedOr
Reds
RedsCP
RedShift
RedTeal
RedYellowBlue
Roma
RomaO
Royal1
Royal2
Royal3

S

Safe
Sapphire
Seasons
Seaweed
Seismic
Sepia
Set1
Set2
Set3
ShiftRainbow
SineBow
Solar
Spectral
Speed
Spring
Summer
SunBurst
Sunset
SunsetDark
Swamp

T

T10
Tab10
Tab20
Tab20b
Tab20c
TableauLight
TableauMedium
TdocBW
TdocCol
TdocDark
Teal
TealGrey
TealGrn
TealRose
Teals
Tempo
Temps
Terrain
Thermal
Tofino
Tokyo
Torch
Toxic
TrafficLight
Tree
Tropic
Tropical
Turbid
Turbo
Turku
Twilight

V

Vanimo
Vik
VikO
Viola
Viridis
Vivid
Voltage

W

Warm
Water
Waterlily
Watermelon
Wildfire
Wind
Winter
Wistia

Y

YellowGrey
YellowPurple
Yellows
YlGn
YlGnBu
YlOrBr
YlOrRd
YlOrRdCBW

Z

Zissou

Appendix 2 – Renamed palettes

Important.

The palette names in this appendix are standard, but some `@prism` implementations add a specific prefix.

To prevent name conflicts, specific suffixes can be applied. Here is the full list of the suffixes used.

Table 1: Suffixes used

CarbonPlan: CP	Colorbrewer: CBW
CARTOColors: CC	Plotly: PLY
CMasher: CM	

Below is the list of all renamed palettes. We use ⇒ to denote these changes, with the new `@prism` names appearing on the right.

Table 2: Renamed palettes

CarbonPlan	Bluegrey	⇒	BlueGrey
	Blues	⇒	BluesCP
	Cool	⇒	CoolCP
	Earth	⇒	EarthCP
	Greengrey	⇒	GreenGrey
	Greens	⇒	GreensCP
	Greys	⇒	GreysCP
	Orangeblue	⇒	OrangeBlue
	Orangegrey	⇒	OrangeGrey
	Oranges	⇒	OrangesCP
	Pinkgreen	⇒	PinkGreen
	Pinkgrey	⇒	PinkGrey
	Purplegrey	⇒	PurpleGrey
	Purples	⇒	PurplesCP
	Rainbow	⇒	RainbowCP
	Redgrey	⇒	RedGrey
	Reds	⇒	RedsCP
	Redteal	⇒	RedTeal
	Sinebow	⇒	SineBow
	Tealgrey	⇒	TealGrey
	Yellowgrey	⇒	YellowGrey
	Yellowpurple	⇒	YellowPurple
CARTOColors	Prism	⇒	PrismCC
CMasher	Copper	⇒	CopperCM
	Fall	⇒	FallCM
	Ghostlight	⇒	GhostLight
	Iceburn	⇒	IceBurn
	Ocean	⇒	OceanCM
	Redshift	⇒	RedShift

Continued on next page

Table 2: Renamed palettes (Continued)

	Sunburst	⇒	SunBurst
Colorbrewer	PRGn	⇒	PrGn
	YlOrRd	⇒	YlOrRdCBW
Matplotlib	Afmhot	⇒	AfmHot
	Brg	⇒	BRG
	Bwr	⇒	BWR
	Coolwarm	⇒	CoolWarm
	Hsv	⇒	HSV
Plotly	Bluered	⇒	BlueRedPLY
	Hot	⇒	HotPLY
	Jet	⇒	JetPLY
	Rainbow	⇒	RainbowPLY
Tableau	Gray	⇒	MidGray

Appendix 3 – Ignored palettes

Important.

The palette names in this appendix are standard, but some `@prism` implementations add a specific prefix.

Excluded by design

The following palettes have been intentionally excluded. Please refer to the explanation below.

Table 3: Excluded palettes

CMasher	`Neutral`	Too big (size > 512).
	`Savanna`	Looks completely black.

Unkept palettes

The following types of palettes are unkept.

- **Exact duplicates:** identical color arrays.
- **Visual identical:** those indistinguishable to the naked eye.[12]
- **Transformations:** reversals, shifts, or a combination of both.

We identify relations between palettes using the following notations.

Duplicates	Visually identical	Reversal	Reversal and shifted
$=$	$\cong$	$\rightleftharpoons$	$\rightleftharpoons$

We list the unkept palettes below, omitting duplicates that share both name and colors. The rightmost palettes are those retained in `@prism`. For instance, the table shows that `Matplotlib`'s `Gray` is the reversal of `@prism`'s `Binary`.

Table 4: Unkept palettes

Matplotlib	`Cubehelix`	$\cong$	`Classic`
	`TwilightShifted`	$\rightleftharpoons$	`Twilight`
	`Gray`	$\rightleftharpoons$	`Binary`
	`GistYarg`	$\rightleftharpoons$	`GistGray`
Plotly	`D3`	$=$	`Tab10`
Tableau	`Tableau`	$=$	`Tab20`

Reconstructing unkept shifted palettes

Let's detail how to retrieve only those excluded palettes from `@prism` that required a shift.

Table 5: Unkept palettes

Matplotlib	`TwilightShifted`	`pal_to_use = Twilight`
		`reverse = True`
		`shift = 256`

[12] In these cases, the original palette is retained. For instance, as `Matplotlib`'s `Cubehelix` is a specific implementation of the original `Cubehelix`'s `Classic`, we keep the latter.

Appendix 4 – Subpalettes

Important.

The palette names in this appendix are standard, but some `@prism` *implementations add a specific prefix.*

For informational purposes, this section lists palettes included within others and describes the process for reconstructing these subpalettes.

Table 6: Unkept palettes

Binary	`pal_to_use = Flag` `indices = 85, 93`
BlueRedPLY	`pal_to_use = BRG` `indices = 0, 1`
BWR	`pal_to_use = Flag` `indices = 73, 85, 97`
Cool	`pal_to_use = ShiftRainbow` `indices = 0, 2`
Speed	`pal_to_use = Delta` `range = from 256 to 511`

Tab10	`pal_to_use = Tab20`
	`range = from 0 to 18`
	`with step 2`

TableauLight	`pal_to_use = Tab20`
	`range = from 1 to 19`
	`with step 2`

Appendix 5 – Palettes by category

Important.

The palette names in this appendix are standard, but some `@prism` implementations add a specific prefix.

Note (Qualitative and semantic palettes).

The threshold of 40 colors has been established for qualitative and semantic palettes. The following lines outline the rationale for this limit.

- *In qualitative color palettes, each color represents a single data point or a group of data. Experience shows that about ten colors are more than enough.*
- *Semantic palettes associate a color with a specific context, such as warning levels (see the frames in this document). Choosing 40 colors allows for 10 different contexts, each with 4 sub-categories. Here are some examples of sub-categories.*

  10-19 Initial / Idle: *default states such as "To Do", "Clickable", or "To be analyzed".*
  20-29 Active / Transition: *states in flux, including "In Progress" or UI "Hover" effects.*
  30-39 Success / Final: *terminal positive states like "Done", "Clicked", or "Validated".*
  40-49 Alert / Rejection: *error handling and negative outcomes, e.g., "Error", "Inactive", or "Rejected".*

Important.

Categories were generated semi-automatically, followed by manual selection to obtain relevant choices.

Colorblind palettes

Colorblind palettes provide high-contrast hues specifically optimized for users with color vision deficiency.

Acton

Apple

Bam

Bamako

BamO

Batlow

BatlowK

BatlowW

Berlin

Bilbao

Broc

BrocO

Buda

Bukavu

Chroma

CopperCM

Cork

CorkO

Davos

Devon

Emergency

Fes

Glasgow

GrayC

Hawaii

Holly

Imola

Lajolla

Lapaz

Lipari

Lisbon

Managua

Navia

NaviaW

Nuuk

OkabeIto

Oleron

Oslo

Petroff6

Petroff8

Petroff10

Pride

Rainforest

Roma

RomaO

Safe

Seasons

Tofino

Tokyo

Turku

Vanimo

Vik

VikO

Watermelon

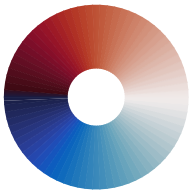
Wildfire

Wistia

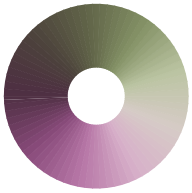
Cyclic palettes

Cyclic palettes create seamless loops for periodic data, such as angles, seasons, or 24-hour cycles.

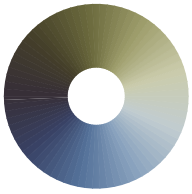
Balance



BamO

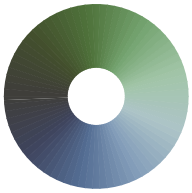


BrocO



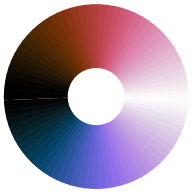
CopperCM

CorkO



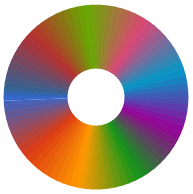
Delta

Emergency

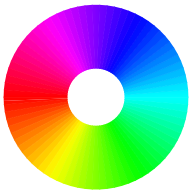


Fusion

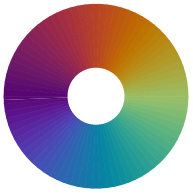
G10



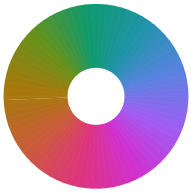
HSV



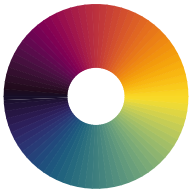
Infinity



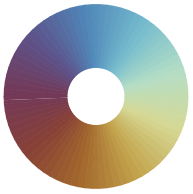
Phase



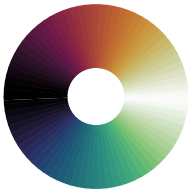
Pride



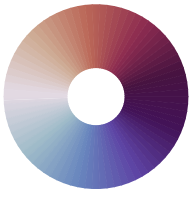
RomaO



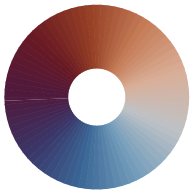
Seasons



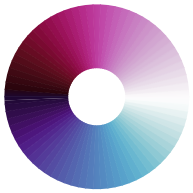
Twilight



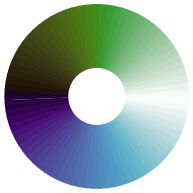
VikO



Viola



Waterlily



Dark palettes

Dark palettes offer high-contrast schemes specifically optimized for terminal interfaces and dark-mode backgrounds.

Autumn

BWR

Cool

GistRainbow

Jet

Lemon

Prism

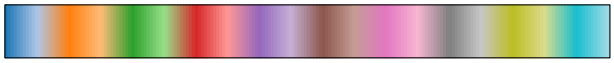
Rainbow

Seismic

Spring

Summer

Tab20



Tab20b

Tab20c

TdocDark

Terrain

Viridis

Winter

YlOrBr

YlOrRd

Divergent palettes

Divergent palettes highlight deviations from one midpoint toward contrasting extremes.

ArmyRose

Balance

Bam

BamO

Batlow

BatlowK

BatlowW

Berlin

Blackbody

BlindFish

BlueDarkOrange12

BlueDarkOrange18

BlueDarkRed12

BlueDarkRed18

BlueGray

BlueGreen

BlueGrey

BlueOrange8

BlueOrange10

BlueOrangeRed

BrBG

Broc

BrocO

BrownBlue10

BrownBlue12

BurningGrass

BWR

CoolWarm

CopperCM

Cork

CorkO

Curl

Delta

Earth

Electric

Emergency

Fall

FantasticFox1

Fusion

Geyser

GrandBudapest1

GrandBudapest2

GreenGrey

GreenMagenta

Guppy

Holly

IceBurn

Lisbon

Managua

OrangeBlue

OrangeGrey

Picnic

PinkGreen

PinkGrey

PiYG

Portland

PrGn

Pride

Prinsenvlag

PuOr

PurpleGrey

RdBu

RdGy

RdYlBu

RdYlGn

RedGrey

RedShift

RedTeal

RedYellowBlue

Roma

Royal1

Seasons

Seaweed

Seismic

Spectral

TealGrey

TealRose

Temps

Tofino

Tropic

Twilight

Vanimo

Vik

VikO

Viola

Waterlily

Watermelon

Wildfire

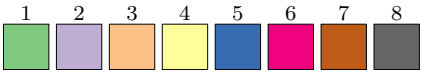
YellowGrey

YellowPurple

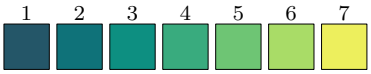
Qualitative palettes

Qualitative palettes use distinct hues to separate categorical data without implying any order or hierarchy.

Accent

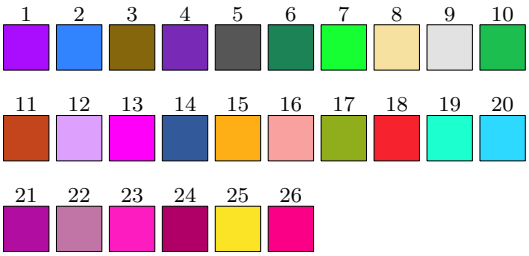


AgGrnYl

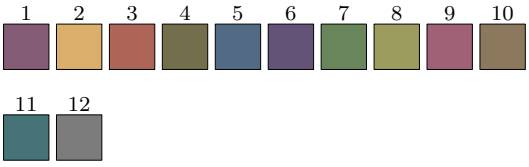


AgSunset

Alphabet

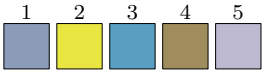


Antique

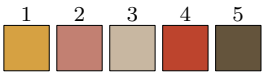


Aquatic1

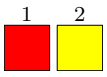
Aquatic2



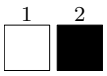
Aquatic3



Autumn

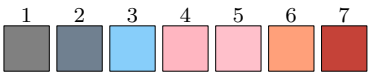


Binary

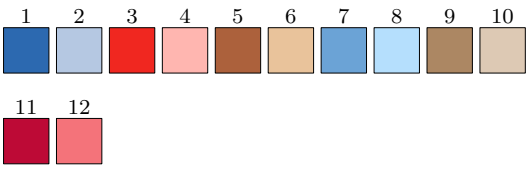


Blackbody

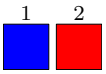
BlindFish



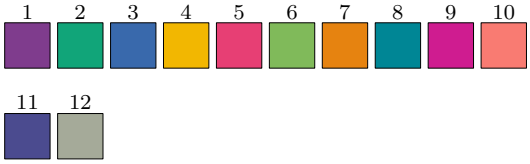
BlueRed



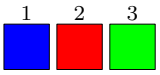
BlueRedPLY



Bold

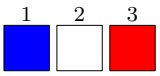


BRG

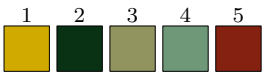


BurningGrass

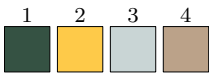
BWR



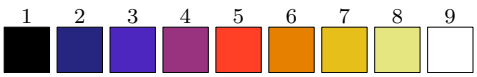
Cavalcanti



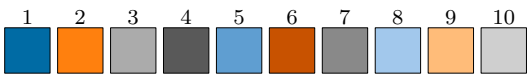
Chevalier



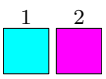
CMRmap



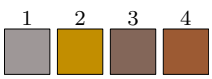
ColorBlind



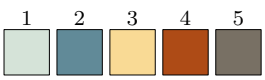
Cool



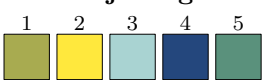
Darjeeling1



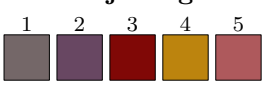
Darjeeling2



Darjeeling3

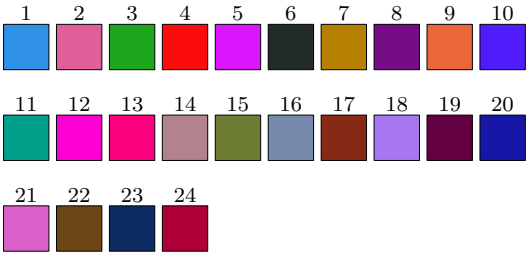


Darjeeling4

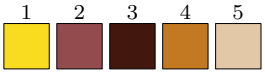


Dark2

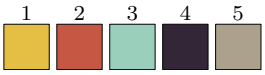
Dark24



FantasticFox1

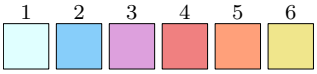


FantasticFox2



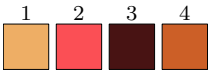
G10

GasFlame

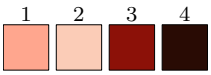


GeoRainbow

GrandBudapest1

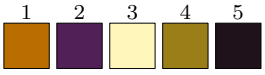


GrandBudapest2

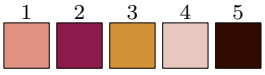


GrandBudapest3

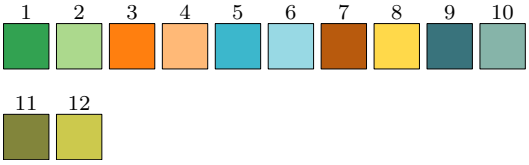
GrandBudapest4



GrandBudapest5



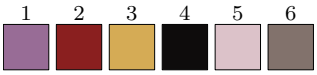
GreenOrange



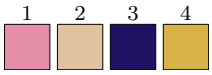
HotPLY

IsleOfDogs1

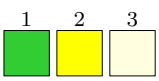
IsleOfDogs2



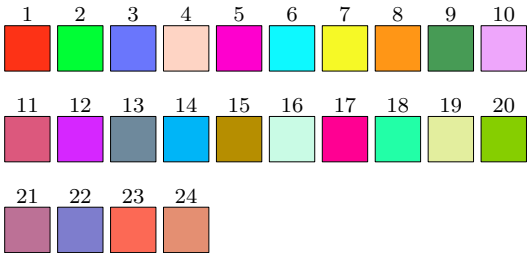
IsleOfDogs3



Lemon

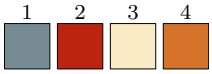


Light24

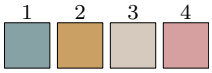


Margot1

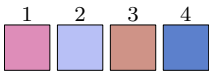
Margot2



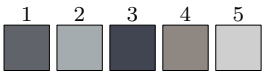
Margot3



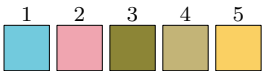
Mendl



MidGray

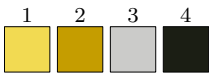


Moonrise1

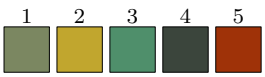


Moonrise2

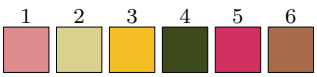
Moonrise3



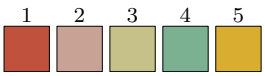
Moonrise4



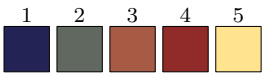
Moonrise5



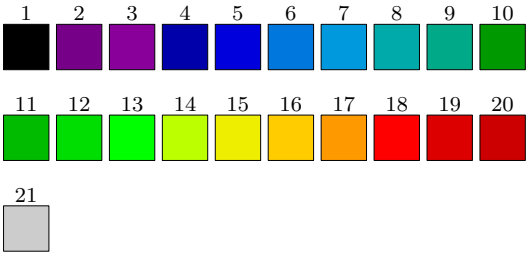
Moonrise6



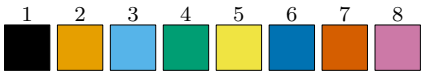
Moonrise7



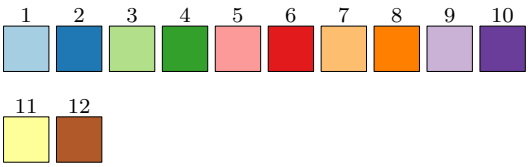
NipySpectral



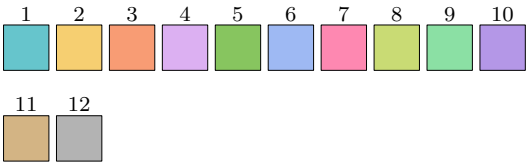
OkabeIto



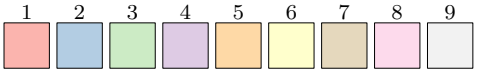
Paired



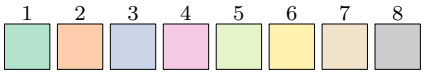
Pastel



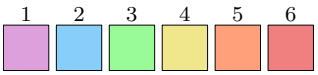
Pastel1



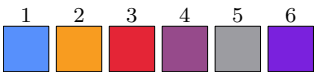
Pastel2



PastelRainbow

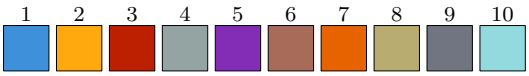


Petroff6

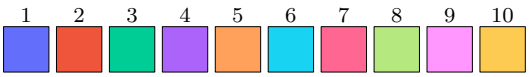


Petroff8

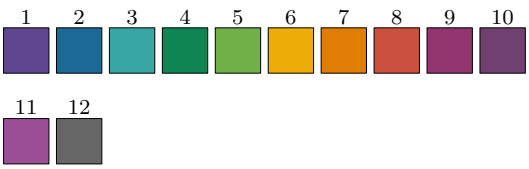
Petroff10



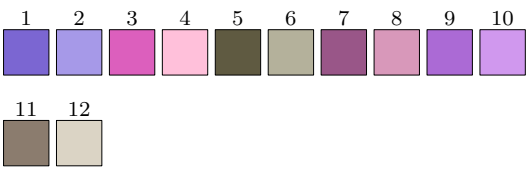
Plotly



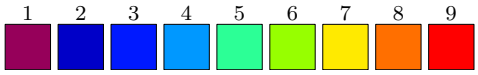
PrismCC



PurpleGray

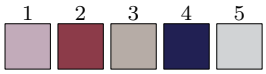


RainbowPLY

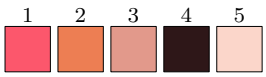


Royal1

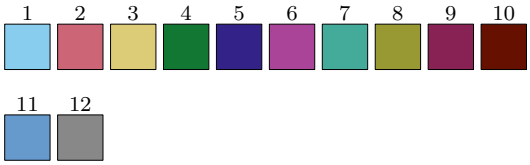
Royal2



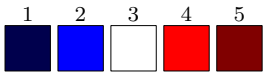
Royal3



Safe

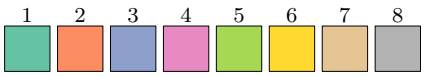


Seismic

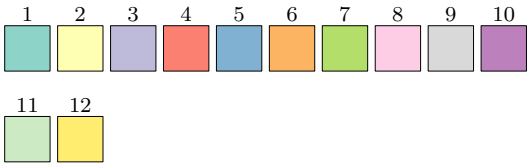


Set1

Set2

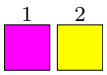


Set3

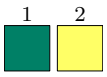


ShiftRainbow

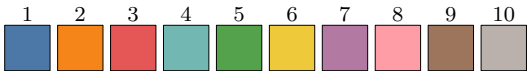
Spring



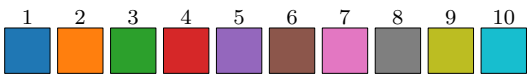
Summer



T10



Tab10



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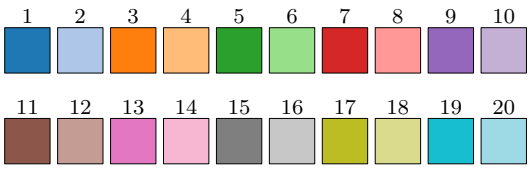
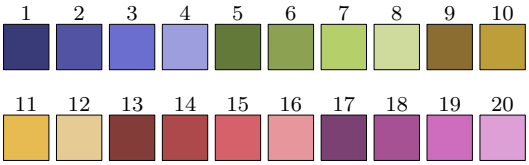
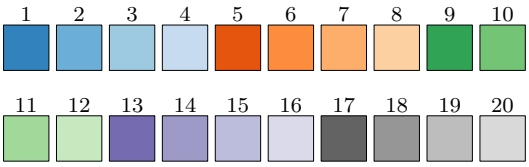


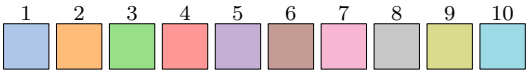
Tab20b



Tab20c



TableauLight

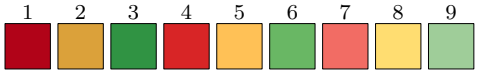


TableauMedium

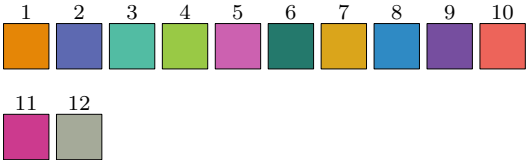
TdocCol

TdocDark

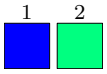
TrafficLight



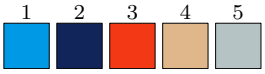
Vivid



Winter

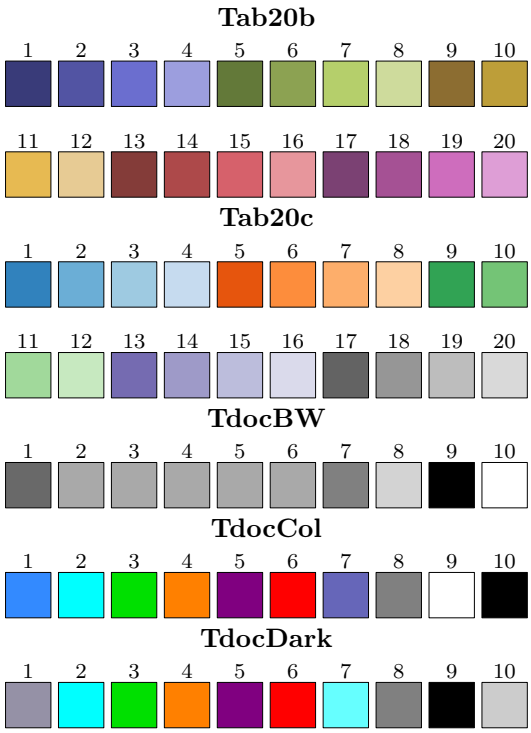


Zissou



Semantic palettes

Semantic palettes are designed to assign colors to categories — such as UI states, system status, or pedagogical elements — to convey immediate meaning.



Sequential palettes

Sequential palettes represent ordered data using a single or multi-hue gradient with monotonic lightness.

Acton

AfmHot

AgGrnYl

AgSunset

Algae

Amber

Amethyst

Amp

Apple

Arctic

ArmyRose

Autumn

Balance

Bam

Bamako

BamO

Batlow

BatlowK

BatlowW

Berlin

Bilbao

Binary

Blackbody

BlindFish

BlueDarkOrange12

BlueDarkOrange18

BlueDarkRed12

BlueDarkRed18

BlueGray

BlueGreen

BlueGrey

BlueOrange8

BlueOrange10

BlueOrangeRed

BlueRedPLY

Blues

Blues7

Blues10

BluesCP

BluGrn

BluYl

Bone

BrBG

BRG

Broc

BrocO

BrownBlue10

BrownBlue12

BrwnYl

Bubblegum

Buda

BuGn

BuPu

Burg

BurgYl

BurningGrass

BWR

Cividis

CMRmap

Cool

CoolCP

CoolWarm

Copper

CopperCM

Cork

CorkO

Cosmic

CubeYF

Curl

DarkMint

Davos

Deep

Delta

Dense

Devon

Earth

EarthCP

Electric

Ember

Emerald

Emergency

Emrld

Fall

FallCM

FantasticFox1

Fire

Flamingo

Freeze

Fusion

GasFlame

Gem

Geyser

GistGray

GistHeat

Glasgow

GnBu

Gnuplot

Gnuplot2

Gothic

GrandBudapest1

GrandBudapest2

Gray

GreenGrey

GreenMagenta

Greens

GreensCP

Greys

GreysCP

Guppy

Haline

Heart

Holly

Horizon

Hot

HotPLY

Ice

IceBurn

Imola

Inferno

JimSpecial

Jungle

Lajolla

Lavender

Lemon

Lipari

Lisbon

Magenta

Magma

Managua

Matter

Mint

Navia

NaviaW

Nuclear

OceanCM

OrangeBlue

OrangeGrey

Oranges

OrangesCP

OrRd

OrYel

Oslo

Peach

Pepper

PerceptualRainbow

Picnic

Pink

PinkGreen

PinkGrey

Pinks

PinkYl

PiYG

Plasma

Portland

PrGn

Pride

Prinsenvlag

PuBu

PuBuGn

PuOr

PuRd

Purp

Purple

PurpleGrey

Purples

PurplesCP

PurpOr

RdBu

RdGy

RdPu

RdYlBu

RdYlGn

Red

RedGrey

RedOr

Reds

RedsCP

RedShift

RedTeal

RedYellowBlue

Roma

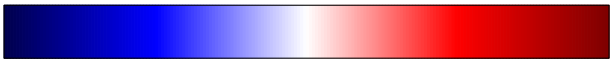
Royal1

Sapphire

Seasons

Seaweed

Seismic



Sepia

Solar

Spectral

Speed

Spring

Summer

SunBurst

Sunset

SunsetDark

Swamp

Teal

TealGrey

TealGrn

TealRose

Teals

Tempo

Temps

Thermal

Tofino

Tokyo

Torch

Toxic

Tree

Tropic

Tropical

Turbid

Turku

Twilight

Vanimo

Vik

VikO

Viola

Viridis

Voltage

Warm

Water

Waterlily

Watermelon

Wildfire

Wind

Winter

Wistia

YellowGrey

YellowPurple

Yellows

YlGn

YlGnBu

YlOrBr

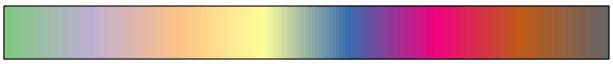
YlOrRd

YlOrRdCBW

Stepped palettes

Stepped palettes are designed to highlight distinct shifts rather than smooth progression. They use high-contrast colors that meet at sharp boundary points to emphasize deviation.

Accent



Antique

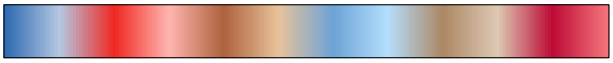
Aquatic1

Aquatic2

Aquatic3

BlueOrange12

BlueRed



Bukavu

Cavalcanti

Chevalier

Chroma

Classic

Cube1

Cubehelix1

Cubehelix2

Cubehelix3

Darjeeling2

Darjeeling3

Darjeeling4

Dark2

Dark24

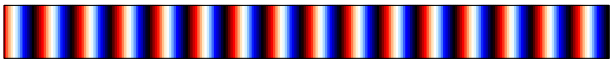
Dusk

Eclipse

FantasticFox2

Fes

Flag



GeoRainbow

GhostLight

GistEarth

GistNcar

GistRainbow

GistStern

GrandBudapest3

GrandBudapest4

GrandBudapest5

Hawaii

HSV

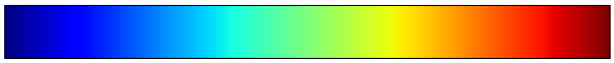
Infinity

IsleOfDogs1

IsleOfDogs2

IsleOfDogs3

Jet



JetPLY

Lapaz

Lilac

LinearL

Margot1

Margot2

Margot3

Mendl

MidGray

Moonrise1

Moonrise2

Moonrise3

Moonrise4

Moonrise5

Moonrise6

Moonrise7

Neon

NipySpectral

Nuuk

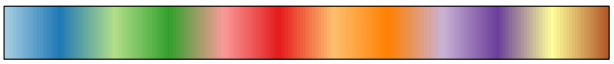
Ocean

OkabeIto

Oleron

Oxy

Paired



Pastel

Pastel1

Pastel2

PastelRainbow

Petroff6

Petroff8

Petroff10

Phase

Prism

PrismCC

PurpleGray

Rainbow

RainbowCP

RainbowPLY

Rainforest

RomaO

Royal2

Royal3

Safe

Set1

Set2

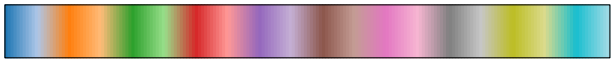
Set3

ShiftRainbow

SineBow

T10

Tab20



Tab20b

Tab20c

TableauLight

TableauMedium

Terrain

TrafficLight

Turbo

Vivid

Zissou